

# **Real-Time Facial Emotion Recognition System Using Machine Learning**

Project Submitted By

Asmita Ghorai

Pallabi Biswas

Ratri Dey

Riddhi Basu

**BACHELOR OF TECHNOLOGY (CSE)**



**DEPARTMENT OF COMPUTER SCIENCE**

Sister Nivedita University, Newtown

Kolkata, West Bengal

2026

# Real-Time Facial Emotion Recognition System Using Machine Learning

Project Submitted in Partial Fulfilment of the Requirements for  
Award of the Degree of

Bachelor of Technology (CSE)

| Student Name   | Reg. Number  | Email ID                    |
|----------------|--------------|-----------------------------|
| Asmita Ghorai  | 220010317574 | ghoraiasmita@gmail.com      |
| Pallabi Biswas | 220010788506 | pallabibiswas4002@gmail.com |
| Ratri Dey      | 220010988119 | ratridey0@gmail.com         |
| Riddhi Basu    | 220011009400 | riddhib4321@gmail.com       |

**Submission Date:** May 8, 2026

*Under the supervision of*

Dr. Bidyut Saha

Assistant Professor

Department of Computer Science

Sister Nivedita University, Newtown

Kolkata, West Bengal



DEPARTMENT OF COMPUTER SCIENCE

Sister Nivedita University, Newtown

Kolkata, West Bengal

May, 2026

## Declaration

*We hereby declare that this dissertation is the product of our own work, and we attest that it contains no material that resulted from collaboration, except where explicitly acknowledged in the text. Furthermore, we confirm that this project has not been previously submitted, either in part or in its entirety, to any other University or Institution for the purpose of obtaining any degree, diploma, or other qualification. All sources used and referenced in this dissertation are duly credited, and any borrowed ideas or information are appropriately cited in accordance with academic standards and guidelines.*

*Date: 08-05-2026*

*Place: SNU, WB*

.....  
*Asmita Ghorai*

*Registration Number: 220010317574*

.....  
*Pallabi Biswas*

*Registration Number: 220010788506*

.....  
*Ratri Dey*

*Registration Number: 220010988119*

.....  
*Riddhi Basu*

*Registration Number: 220011009400*



## *Certificate*

*This is to certify that the project entitled "Real-Time Facial Emotion Recognition System Using Machine Learning", submitted by Asmita Ghorai , Pallabi Biswas, Ratri Dey, Riddhi Basu to Sister Nivedita University, West Bengal for the award of the degree of Bachelor of Technology (CSE) is a bonafide record of the project work carried out by them under my supervision and guidance. The content of the project, in full or parts have not been submitted to any other institute or university for the award of any degree or diploma.*

*Date: 08-05-2026  
Place: SNU, WB*

.....  
*(Dr. Bidyut Saha)  
Assistant Professor  
Dept. of Computer Science  
Sister Nivedita University*

# Contents

|                                                          |             |
|----------------------------------------------------------|-------------|
| <b>List of Tables</b>                                    | <b>vii</b>  |
| <b>List of Figures</b>                                   | <b>viii</b> |
| <b>Acknowledgement</b>                                   | <b>ix</b>   |
| <b>1 Introduction</b>                                    | <b>1</b>    |
| 1.1 Background / Context of the Study . . . . .          | 1           |
| 1.2 Problem Context / Motivation . . . . .               | 1           |
| 1.3 Problem Statement . . . . .                          | 1           |
| 1.4 Objectives of the Project . . . . .                  | 2           |
| 1.5 Scope of the Project . . . . .                       | 2           |
| 1.6 Significance / Contribution . . . . .                | 2           |
| 1.7 Methodology Overview . . . . .                       | 3           |
| 1.8 Organization of the Report . . . . .                 | 3           |
| <b>2 Literature Survey</b>                               | <b>4</b>    |
| 2.1 Introduction to Literature Survey . . . . .          | 4           |
| 2.2 Existing Systems / Related Work . . . . .            | 4           |
| 2.3 Limitations of Existing Systems . . . . .            | 5           |
| 2.4 Comparative Study / Gap Analysis . . . . .           | 5           |
| 2.5 Summary of Literature Review . . . . .               | 5           |
| <b>3 Problem Identification</b>                          | <b>7</b>    |
| 3.1 Context / Background of the Problem . . . . .        | 7           |
| 3.2 Problem Definition . . . . .                         | 7           |
| 3.3 Problem Environment / Current Scenario . . . . .     | 7           |
| 3.4 Need / Importance of Solving the Problem . . . . .   | 8           |
| 3.5 Scope and Boundaries of the Problem . . . . .        | 8           |
| 3.6 Objectives (High-level) . . . . .                    | 8           |
| 3.7 Expected Challenges . . . . .                        | 8           |
| <b>4 Proposed Methodology</b>                            | <b>10</b>   |
| 4.1 Introduction to Methodology . . . . .                | 10          |
| 4.2 System Architecture / Conceptual Framework . . . . . | 10          |
| 4.3 Algorithms / Techniques Used . . . . .               | 13          |
| 4.4 Model Architecture Description . . . . .             | 14          |
| 4.5 Data Handling and Processing . . . . .               | 14          |
| 4.6 Workflow of the Proposed System . . . . .            | 15          |
| 4.7 Expected Outcomes . . . . .                          | 15          |
| 4.8 Novelty / Uniqueness of the Methodology . . . . .    | 16          |
| <b>5 Results and Discussion</b>                          | <b>17</b>   |
| 5.1 Hardware and Software Environments . . . . .         | 17          |
| 5.2 Databases Used . . . . .                             | 17          |

|          |                                                             |           |
|----------|-------------------------------------------------------------|-----------|
| 5.3      | Experimental Results and Discussion . . . . .               | 18        |
| 5.3.1    | Performance Metrics . . . . .                               | 18        |
| 5.3.2    | Accuracy and Loss Analysis . . . . .                        | 19        |
| 5.3.3    | Confusion Matrix Analysis . . . . .                         | 20        |
| 5.3.4    | Comparison with Existing State-of-the-Art Methods . . . . . | 21        |
| 5.3.5    | Computational Complexity Analysis . . . . .                 | 21        |
| 5.3.6    | Novelty of the Proposed Methodology . . . . .               | 21        |
| <b>6</b> | <b>Conclusion and Future Work</b>                           | <b>22</b> |
| 6.1      | Summary of Work . . . . .                                   | 22        |
| 6.2      | Key Contributions . . . . .                                 | 22        |
| 6.3      | Limitations of the Project . . . . .                        | 23        |
| 6.4      | Future Scope / Extensions . . . . .                         | 23        |
|          | <b>References</b>                                           | <b>24</b> |

# List of Tables

|                                                                  |   |
|------------------------------------------------------------------|---|
| 2.1 Comparison of Existing Emotion Recognition Methods . . . . . | 5 |
|------------------------------------------------------------------|---|

# List of Figures

|     |                                                            |    |
|-----|------------------------------------------------------------|----|
| 4.1 | Workflow of Data Processing . . . . .                      | 11 |
| 4.2 | Workflow of Preprocessing and Model Architecture . . . . . | 12 |
| 4.3 | Workflow of Training and Evaluation . . . . .              | 13 |
| 5.1 | Summary of Model Performance Metrics . . . . .             | 18 |
| 5.2 | Training and Validation Accuracy Graph . . . . .           | 19 |
| 5.3 | Training and Validation Loss Graph . . . . .               | 20 |
| 5.4 | Confusion Matrix of Emotion Classification . . . . .       | 21 |



# Acknowledgement

We would like to express our heartfelt gratitude to our esteemed guide, **Dr. Bidyut Saha**, Assistant Professor at Sister Nivedita University, WB, for his invaluable guidance, unwavering support, and encouragement throughout our project journey. His profound knowledge, insightful suggestions, and mentorship have been instrumental in shaping our project and academic pursuits.

We extend our sincere thanks to the members of the Project Committee at the Department of Computer Science, Sister Nivedita University, for their valuable feedback and constructive criticism during the evaluation of our work. Their expertise and insights have been crucial in refining the quality of our project.

We would like to express our heartfelt gratitude to our HOD, the esteemed Vice-Chancellor, the Class teachers and all the faculty members of our department for their unwavering support and guidance throughout our project journey. Your invaluable contributions have been instrumental in shaping this project endeavor.

We are deeply indebted to our family members whose constant love, encouragement, and support have been the driving force behind our academic pursuits. Their unwavering belief in us has given us the strength to overcome challenges and achieve milestones.

We also want to express our heartfelt gratitude to our friends, colleagues, juniors, and seniors for their camaraderie, encouragement, and support throughout this academic journey. Their presence and interactions have enriched us with diverse perspectives, making this journey all the more rewarding.

Lastly, we extend our sincere appreciation to all those who have been our pillars of support during this project endeavor. Your encouragement, motivation, and belief in our abilities have been invaluable, and we are truly grateful for your presence in our lives.

Asmita Ghorai  
(Registration No.: 220010317574) .....

Pallabi Biswas  
(Registration No.: 220010788506) .....

Ratri Dey  
(Registration No.: 220010988119) .....

Riddhi Basu  
(Registration No.: 220011009400) .....

# Abstract

The project titled “**Real-Time Facial Emotion Recognition System using Machine Learning**” aims to analyze and classify human emotional states using facial expressions in real time. Traditional methods of emotion analysis rely heavily on manual observation, which is subjective, time-consuming, and often inefficient. To address this limitation, the proposed system employs a machine learning-based Convolutional Neural Network (CNN) for efficient and accurate facial emotion recognition. The system processes input images through face detection using OpenCV Haar Cascade, followed by preprocessing techniques such as cropping, resizing, and normalization. The model is trained on a dataset consisting of seven emotion classes: angry, disgust, fear, happy, neutral, sad, and surprise. Performance optimization techniques including data augmentation, early stopping, and learning rate reduction are applied to enhance generalization and prevent overfitting. The system is evaluated using accuracy metrics, top-k accuracy, and confusion matrix analysis, achieving a final training accuracy of 91.23% and validation accuracy of 84.0%. Furthermore, a real-time webcam-based implementation enables live emotion detection, making the system practical and applicable in various real-world scenarios. The proposed system provides an automated, scalable, and objective solution for emotion analysis, with potential applications in human-computer interaction, healthcare monitoring, security systems, and intelligent user experience design, while future enhancements may include improved model accuracy, multi-face detection, and integration with advanced analytics systems.

**Keywords:** Machine Learning, Emotion Recognition, Convolutional Neural Network, OpenCV, Real-Time System, Computer Vision

# Chapter 1

## Introduction

### 1.1 Background / Context of the Study

In recent years, the integration of machine learning and computer vision has significantly enhanced the ability to analyze human behavior and emotional states. With the increasing adoption of intelligent systems across various domains such as education, healthcare, security, and human-computer interaction, there is a growing need for automated solutions that can provide real-time insights into human emotions. Traditional observation-based methods are often subjective, time-consuming, and inefficient for continuous monitoring.

Facial expressions serve as a key indicator of human emotions, engagement levels, and cognitive responses. Advances in machine learning-based Convolutional Neural Networks (CNNs) have made it possible to effectively detect and classify emotions from facial images. This project leverages these advancements to develop a real-time emotion recognition system capable of classifying facial expressions into seven categories: angry, disgust, fear, happy, neutral, sad, and surprise.

### 1.2 Problem Context / Motivation

In real-world scenarios, it is challenging to continuously monitor emotional states manually due to limitations in human observation and scalability. Individuals experiencing confusion, stress, or disengagement may not explicitly communicate their emotional state, making it difficult to respond effectively in dynamic environments.

With the advancement of machine learning techniques, there is a strong motivation to automate emotion detection and analysis. This creates a gap between the need for continuous emotional analysis and the limitations of existing manual or semi-automated approaches. A real-time emotion recognition system can assist in multiple domains by enabling better decision-making, improving interaction quality, and enhancing user experience.

### 1.3 Problem Statement

The problem addressed in this project is the lack of an efficient, scalable, and automated system for real-time facial emotion recognition. Existing approaches often rely on manual observation or lack the capability to perform real-time analysis under varying conditions. The proposed system aims to classify facial expressions into seven emotion categories using a

machine learning-based CNN model integrated with real-time webcam input.

## 1.4 Objectives of the Project

The primary objectives of this project are:

- To develop a machine learning-based system for real-time emotion recognition.
- To classify facial expressions into seven categories: angry, disgust, fear, happy, neutral, sad, and surprise.
- To implement face detection using OpenCV and preprocess images through cropping, resizing, and normalization.
- To design and implement a custom Convolutional Neural Network (CNN) for emotion classification.
- To apply data augmentation and training optimization techniques to enhance model generalization.
- To evaluate the system using accuracy metrics, top-k accuracy, and confusion matrix analysis.
- To implement a real-time webcam-based emotion detection system.

## 1.5 Scope of the Project

This project focuses on facial emotion recognition using machine learning techniques. The scope includes dataset preprocessing, model training, evaluation, and real-time implementation using webcam input. The system is designed to analyze facial expressions and does not consider other behavioral cues such as voice, body posture, or eye tracking. Additionally, large-scale deployment and integration with enterprise-level systems are beyond the scope of this work.

## 1.6 Significance / Contribution

The proposed system contributes to the development of intelligent systems by providing an automated and objective method for analyzing human emotions. It reduces reliance on manual observation and enables continuous monitoring in real-time environments. The integration of machine learning with real-time emotion detection enhances practical applicability across multiple domains. Furthermore, the use of a custom CNN architecture ensures a balance between accuracy and computational efficiency.

## **1.7 Methodology Overview**

The proposed Real-Time Facial Emotion Recognition System follows a systematic methodology starting with the collection of a labeled dataset containing seven emotion classes. The images are preprocessed by resizing and normalizing, and data augmentation techniques are applied to improve generalization. A custom Convolutional Neural Network (CNN) model is then designed with multiple convolutional and pooling layers for feature extraction, followed by fully connected layers for classification. The model is trained using the Adam optimizer along with callbacks such as EarlyStopping, ModelCheckpoint, and ReduceLROnPlateau to enhance performance. After training, the model is evaluated using metrics like accuracy, top-k accuracy, and confusion matrix analysis to assess its effectiveness. The best-performing model is saved as an .h5 file. Finally, the system is deployed locally using OpenCV, where real-time facial emotion prediction is performed through a webcam feed.

## **1.8 Organization of the Report**

This dissertation is organized as follows. Chapter 2 presents a detailed literature survey on facial emotion recognition and machine learning techniques. Chapter 3 defines the problem, discusses existing challenges, and outlines the objectives of the project. Chapter 4 describes the proposed methodology, including system architecture, algorithms, and workflow. Chapter 5 presents experimental results, performance evaluation, and analysis of the proposed system. Chapter 6 concludes the work by summarizing key findings and suggesting future enhancements.

# Chapter 2

## Literature Survey

### 2.1 Introduction to Literature Survey

In recent years, facial emotion recognition has gained significant attention in the fields of machine learning and computer vision due to its wide range of applications in human-computer interaction, surveillance, healthcare, and intelligent systems. The ability to automatically detect and classify human emotions from facial expressions has enabled the development of systems capable of understanding human behavior.

Several approaches have been proposed for emotion recognition, ranging from traditional machine learning techniques using handcrafted features to advanced neural network-based models. This chapter presents a review of existing works related to facial emotion recognition, highlighting their methodologies, performance, and limitations.

### 2.2 Existing Systems / Related Work

Early approaches to facial emotion recognition relied on feature extraction techniques such as Local Binary Patterns (LBP), Histogram of Oriented Gradients (HOG), and Scale-Invariant Feature Transform (SIFT). These methods required manual feature engineering and were often limited in handling variations in lighting, pose, and facial expressions.

With advancements in machine learning, neural network-based models such as Convolutional Neural Networks (CNNs) have significantly improved emotion recognition accuracy. Researchers have utilized architectures such as VGGNet, ResNet, and MobileNet for feature extraction and classification. MobileNetV2, in particular, is widely used due to its lightweight architecture and efficiency, making it suitable for real-time applications.

Several studies have also explored transfer learning techniques, where pre-trained models are fine-tuned on emotion datasets such as FER-2013 and CK+. These approaches reduce training time and improve performance by leveraging previously learned features. Additionally, real-time emotion recognition systems using OpenCV and webcam input have been developed to enable live monitoring and analysis.

## 2.3 Limitations of Existing Systems

Despite significant advancements, existing systems face several challenges:

- Many existing systems perform well on controlled datasets but fail in real-time scenarios due to variations in lighting, background, and camera quality.
- Most models are trained on limited or imbalanced datasets, which reduces their ability to generalize across different age groups, ethnicities, and environments.
- Some systems focus only on static image classification and lack real-time implementation capabilities.
- Accuracy may decrease due to variations in lighting, occlusion, and facial orientation.
- Limited adaptability across diverse real-world applications and environments.
- Advanced deep learning models often require high computational power, making real-time deployment difficult on low-end devices.

## 2.4 Comparative Study / Gap Analysis

A comparative analysis of selected existing methods is presented in Table 2.1. The table highlights the techniques used, datasets, performance, and limitations of each approach.

Table 2.1: Comparison of Existing Emotion Recognition Methods

| Method          | Technique                             | Dataset                 | Performance           | Limitations                   |
|-----------------|---------------------------------------|-------------------------|-----------------------|-------------------------------|
| LBP-based Model | Handcrafted Features                  | CK+                     | Moderate Accuracy     | Sensitive to lighting         |
| HOG + SVM       | Feature-based ML                      | FER-2013                | Moderate Accuracy     | Limited generalization        |
| VGGNet          | CNN-based Model                       | FER-2013                | High Accuracy         | High computational cost       |
| ResNet          | CNN-based Model                       | FER-2013                | High Accuracy         | Complex architecture          |
| MobileNetV2     | Transfer Learning (CNN)               | FER-2013                | High Accuracy         | Slight performance trade-off  |
| Proposed System | CNN-based Model (Custom Architecture) | FER-2013 (Preprocessed) | Satisfactory Accuracy | Requires further optimization |

From the above analysis, it is evident that while machine learning-based approaches provide improved accuracy compared to traditional methods, there is still a need for systems that balance performance, efficiency, and real-time capability. This gap motivates the development of the proposed system.

## 2.5 Summary of Literature Review

This chapter reviewed various techniques used for facial emotion recognition, including traditional machine learning approaches and modern neural network-based models. While these methods have demonstrated strong performance, challenges such as computational complexity and limited real-time implementation remain. The analysis highlights the need for an efficient and scalable

system capable of real-time emotion detection across diverse application domains, which forms the foundation of the proposed work.



# Chapter 3

## Problem Identification

### 3.1 Context / Background of the Problem

In modern real-world environments, understanding human emotions has become increasingly important across various domains such as healthcare, security, human-computer interaction, and intelligent systems. The ability to analyze emotional states can significantly improve decision-making, user experience, and system responsiveness. Traditional methods of emotion analysis rely on manual observation, which is often subjective, inconsistent, and inefficient for continuous monitoring.

Machine learning and computer vision technologies have enabled automated analysis of facial expressions, making it possible to detect emotions from visual data. However, implementing such systems effectively in practical scenarios remains a challenge due to variations in environmental conditions and system limitations.

### 3.2 Problem Definition

The primary problem addressed in this project is the lack of an efficient, scalable, and automated system for facial emotion recognition. Existing approaches often rely on manual observation or fail to provide continuous and real-time analysis under varying conditions.

This project aims to develop a machine learning-based system that can effectively classify human facial expressions into multiple emotion categories, thereby providing objective insights into emotional states.

### 3.3 Problem Environment / Current Scenario

Currently, emotion analysis is often performed using indirect or offline methods that do not provide continuous insights. Many existing systems operate on static images and lack real-time implementation capabilities.

Additionally, several emotion recognition models require high computational resources or perform inconsistently under varying lighting conditions, facial orientations, and occlusions. These limitations reduce their effectiveness in real-world environments.

### 3.4 Need / Importance of Solving the Problem

There is a strong need for an automated system capable of continuously monitoring and analyzing human emotions. Such a system can enhance interaction quality, improve system responsiveness, and support decision-making across various applications.

The development of a real-time emotion recognition system can benefit multiple domains, including healthcare monitoring, security systems, user behavior analysis, and human-computer interaction. It provides an objective and scalable approach to understanding emotional states and bridges the gap between manual observation and intelligent automated systems.

### 3.5 Scope and Boundaries of the Problem

This project focuses on facial emotion recognition using machine learning techniques. The system analyzes facial expressions captured through a webcam and classifies them into predefined emotion categories.

However, the scope is limited to visual facial data and does not include other behavioral indicators such as voice tone, body language, or eye movement. The system is designed for controlled environments and does not address large-scale deployment or integration with enterprise-level systems.

### 3.6 Objectives (High-level)

The high-level objectives of this project are:

- To design a machine learning-based emotion recognition system.
- To enable real-time detection of human emotions using webcam input.
- To improve the accuracy and efficiency of emotion classification.
- To provide meaningful insights into emotional states.

### 3.7 Expected Challenges

The development of the proposed system involves several challenges:

- Variations in lighting conditions and image quality.
- Differences in facial orientation and occlusion.
- Limited availability of high-quality labeled datasets.

- Computational constraints for real-time processing.
- Ensuring consistent accuracy across diverse users and environments.

# Chapter 4

## Proposed Methodology

### 4.1 Introduction to Methodology

The proposed system aims to develop a machine learning-based real-time emotion recognition system for analyzing human emotional states in real-world environments. A structured methodology is required to ensure efficient data processing, model training, and accurate emotion classification.

The system follows a machine learning-based approach using computer vision techniques. It integrates image preprocessing, feature extraction, and classification to detect and categorize facial expressions into seven emotion classes.

### 4.2 System Architecture / Conceptual Framework

The overall system architecture is divided into multiple layers, each responsible for a specific stage in the emotion recognition pipeline. The architecture ensures efficient data processing, model training, evaluation, and real-time deployment.

- **Data Collection Layer:** This layer is responsible for collecting facial image datasets containing seven emotion classes (Angry, Disgust, Fear, Happy, Neutral, Sad, Surprise), which are used for training and validation.
- **Preprocessing Layer:** In this stage, images undergo face detection, cropping, resizing to a fixed size (96×96), and normalization. Data augmentation techniques such as rotation, flipping, and zooming are also applied to enhance dataset diversity and reduce overfitting.
- **Feature Extraction Layer:** This layer uses multiple Convolutional Neural Network (CNN) layers to automatically extract important facial features such as edges, textures, and expression patterns.
- **Training Layer:** The extracted features are used to train the CNN model using the Adam optimizer and sparse categorical cross-entropy loss. Callbacks such as EarlyStopping and ReduceLROnPlateau are used to improve training efficiency and prevent overfitting.
- **Evaluation Layer:** The trained model is evaluated using performance metrics such as accuracy, top-k accuracy, and confusion matrix to analyze prediction performance and identify misclassifications.

- **Model Storage Layer:** The best-performing model is saved as `best_model.h5` for future use and deployment.
- **Application / Deployment Layer:** The saved model is integrated with OpenCV in a local environment (e.g., VS Code). The system captures real-time video through a webcam, detects faces, and predicts emotions instantly, displaying the results on the screen.

The data flows sequentially through these layers, starting from data collection and preprocessing, followed by feature extraction, model training, evaluation, and final deployment for real-time emotion recognition.

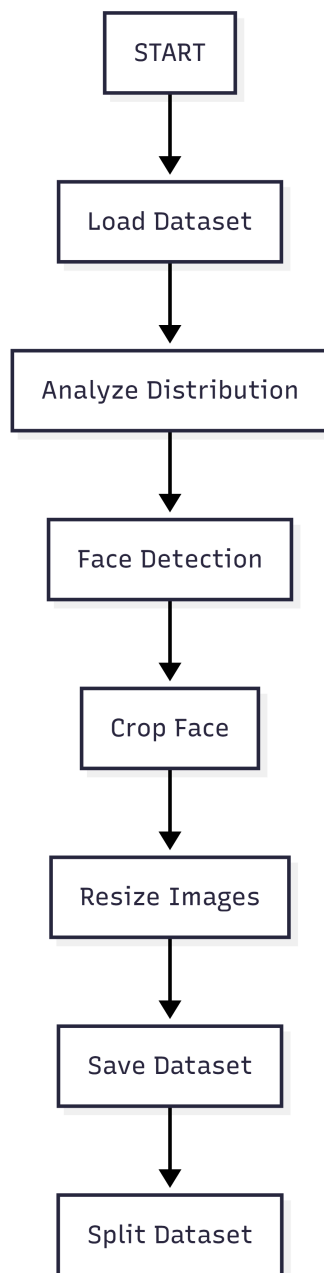


Figure 4.1: Workflow of Data Processing

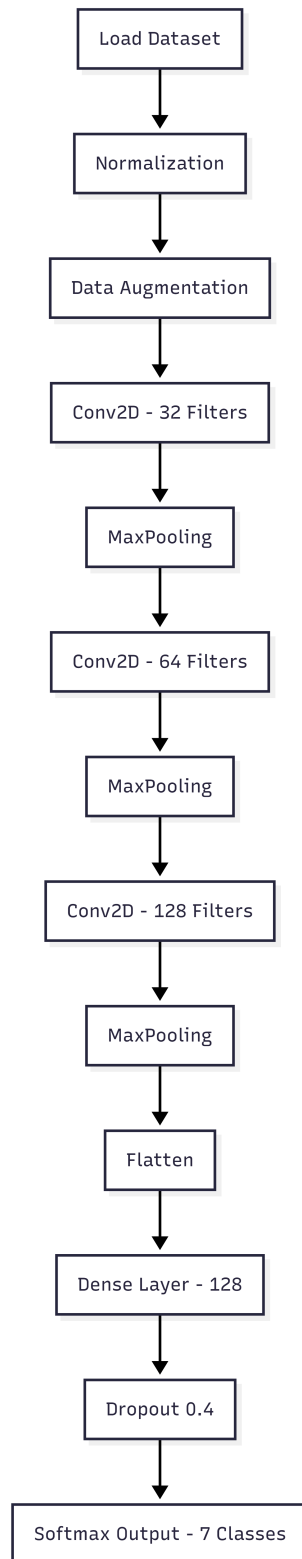


Figure 4.2: Workflow of Preprocessing and Model Architecture

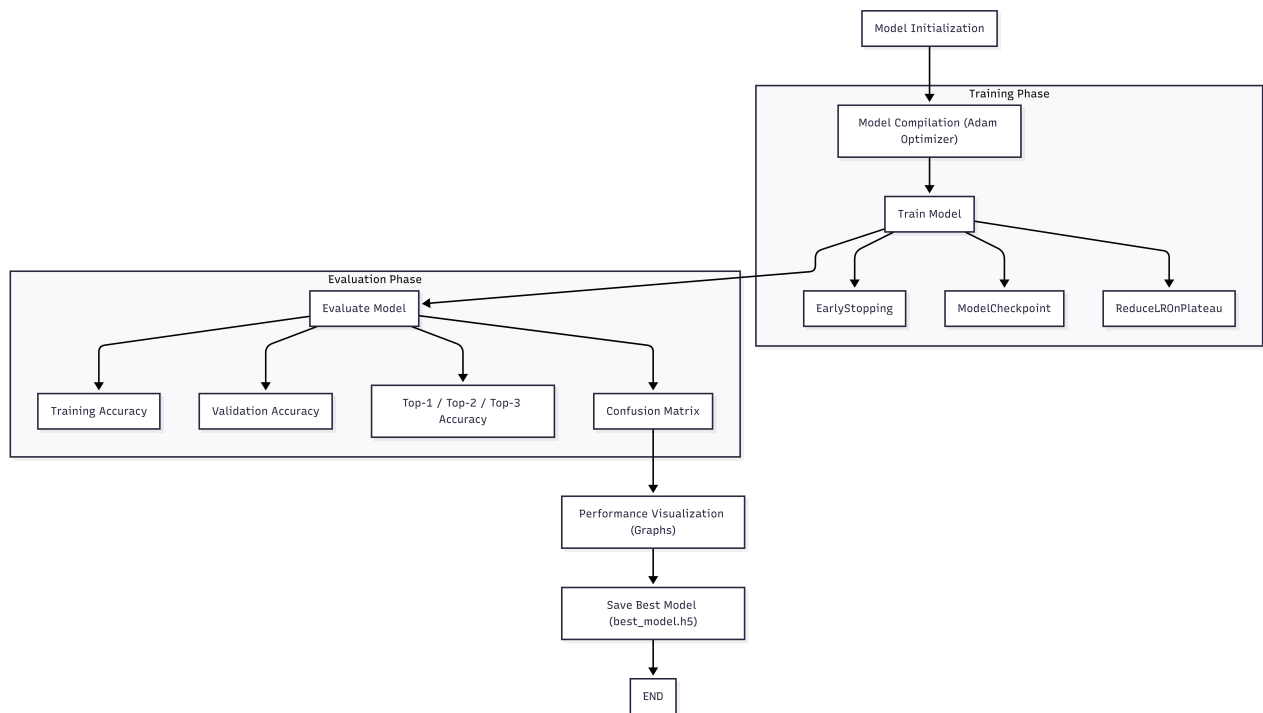


Figure 4.3: Workflow of Training and Evaluation

### 4.3 Algorithms / Techniques Used

The proposed system utilizes the following techniques:

- **Face Detection:** Haar Cascade classifier from OpenCV is used to detect faces in images and video frames.
- **Feature Extraction and Classification:** A Convolutional Neural Network (CNN) is used to automatically extract features from facial images and perform emotion classification.
- **Optimization Technique:** The Adam optimizer is used to achieve faster convergence and improved model performance during training.
- **Regularization:** Dropout is applied within the network to reduce overfitting and improve model generalization.
- **Training Enhancement:** Techniques such as EarlyStopping, ModelCheckpoint, and ReduceLRonPlateau are used to optimize training, prevent overfitting, and improve model stability.

The CNN architecture is designed to balance accuracy and computational efficiency, making it suitable for real-time applications.

## 4.4 Model Architecture Description

The proposed system utilizes a Custom Convolutional Neural Network (CNN) architecture designed specifically for facial emotion recognition. The model is built using a sequential architecture consisting of multiple convolutional and pooling layers followed by fully connected layers for classification.

The input layer accepts RGB images of size  $96 \times 96 \times 3$ , which are normalized to ensure consistent input distribution. Data augmentation techniques such as horizontal flipping and slight rotation are applied during preprocessing to improve model generalization and reduce overfitting.

The CNN architecture consists of three convolutional blocks. Each block includes a Conv2D layer with ReLU activation followed by a MaxPooling layer. The convolutional layers use 32, 64, and 128 filters respectively, enabling the model to progressively extract hierarchical features from the input images, starting from low-level features such as edges and textures to higher-level facial patterns.

After feature extraction, the resulting feature maps are passed through a Flatten layer, which converts them into a one-dimensional feature vector. This is followed by a fully connected Dense layer with 128 neurons and ReLU activation to learn complex patterns from the extracted features. A Dropout layer with a dropout rate of 0.4 is applied to reduce overfitting and improve model generalization.

The final output layer consists of a Dense layer with 7 neurons and a Softmax activation function, which produces probability scores for each of the emotion classes: Angry, Disgust, Fear, Happy, Neutral, Sad, and Surprise.

The model is compiled using the Adam optimizer with a learning rate of 0.0003 and trained using sparse categorical cross-entropy loss. Accuracy is used as the primary evaluation metric. The architecture is designed to balance performance and computational efficiency, making it suitable for real-time emotion recognition applications.

## 4.5 Data Handling and Processing

The dataset consists of facial images categorized into seven emotion classes: angry, disgust, fear, happy, neutral, sad, and surprise.

The data processing steps include:

- Face detection using Haar Cascade to identify facial regions in images.
- Cropping the detected face regions to focus on relevant features.
- Resizing images to a fixed dimension of  $96 \times 96$  pixels to match the model input size.



- Normalizing pixel values to the range [0,1] to ensure consistent input distribution and improve model performance.
- Applying data augmentation techniques such as rotation, zooming, and horizontal flipping to increase dataset diversity and reduce overfitting.
- Loading images using directory-based batching to enable efficient data processing and training.

The dataset is divided into training and validation sets in an approximate 80:20 ratio to ensure proper model evaluation and generalization.

## 4.6 Workflow of the Proposed System

The proposed system follows a sequential workflow for real-time facial emotion recognition:

- Input is captured as a facial image or live video frame through a webcam.
- Faces are detected using the Haar Cascade classifier from OpenCV.
- The detected facial region is cropped and resized to  $96 \times 96$  pixels to match the model input size.
- The image is normalized by scaling pixel values to the range [0,1].
- Relevant features are extracted using the convolutional layers of the CNN model.
- The trained model classifies the input into one of the seven emotion categories: angry, disgust, fear, happy, neutral, sad, and surprise.
- The predicted emotion is displayed in real time on the video frame.

## 4.7 Expected Outcomes

The proposed system is expected to achieve:

- Accurate classification of facial expressions.
- Real-time emotion detection using webcam input.
- Efficient performance using a lightweight CNN architecture.
- Scalable solution applicable across various real-world domains.
- Reliable performance under varying real-world conditions.

## **4.8 Novelty / Uniqueness of the Methodology**

The novelty of the proposed methodology lies in the design of a structured and efficient pipeline for real-time facial emotion recognition using deep learning techniques. The system integrates key stages including face detection, image preprocessing, feature extraction using a custom Convolutional Neural Network (CNN), and real-time prediction.

Unlike traditional approaches that rely on manual observation or offline analysis, the proposed methodology ensures continuous and automated emotion detection through a streamlined workflow. The use of a lightweight custom CNN architecture allows effective feature extraction while maintaining computational efficiency.

Furthermore, the integration of preprocessing techniques, training optimization methods, and real-time webcam-based input enhances the practicality and adaptability of the system. This combination of components results in a scalable and efficient methodology suitable for real-world deployment.

# Chapter 5

## Results and Discussion

### 5.1 Hardware and Software Environments

The implementation of the proposed system was carried out using a standard computing environment. The tools, technologies, and platforms used are as follows:

#### **Software Tools:**

- Programming Language: Python
- Development Environment: Jupyter Notebook / Google Colab

#### **Frameworks and Libraries:**

- TensorFlow and Keras for model development
- OpenCV for image processing and face detection
- NumPy for numerical computations
- Matplotlib for visualization

#### **Hardware:**

- Processor: Intel Core i5/i7 or equivalent
- RAM: 8 GB or higher
- Storage: Sufficient disk space for dataset and model
- Webcam for real-time emotion detection

### 5.2 Databases Used

The dataset used for this project was obtained from Kaggle (FER-2013 dataset) and consists of facial images categorized into seven emotion classes: angry, disgust, fear, happy, neutral, sad, and surprise.

The dataset was preprocessed using OpenCV Haar Cascade for face detection, followed by cropping, resizing, and normalization. The processed dataset was divided into training and validation sets in an 80:20 ratio. Data augmentation techniques such as flipping and rotation were applied to improve model generalization.

## 5.3 Experimental Results and Discussion

### 5.3.1 Performance Metrics

The performance of the model is evaluated using various metrics such as accuracy, top-1 accuracy, top-2 accuracy, top-3 accuracy, and confusion matrix analysis. These metrics provide a comprehensive understanding of the model's prediction capability and reliability.

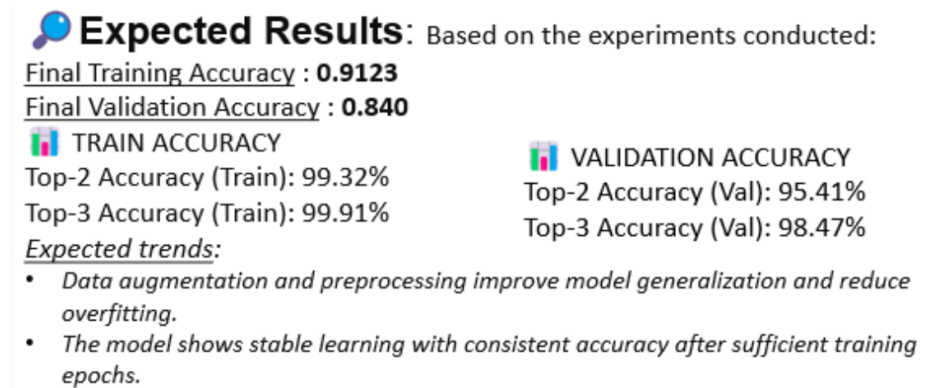


Figure 5.1: Summary of Model Performance Metrics

The model achieved a final training accuracy of 91.23% and validation accuracy of 84.0%. The top-1 accuracy differs slightly from the final epoch accuracy due to evaluation across different stages of the training process. The gap between training and validation accuracy indicates slight overfitting, which is controlled through data augmentation and regularization techniques.

#### Detailed Accuracy Metrics:

##### ■ Training Accuracy:

- Top-1 Accuracy: 94.17%
- Top-2 Accuracy: 99.32%
- Top-3 Accuracy: 99.91%

##### ■ Validation Accuracy:

- Top-1 Accuracy: 83.67%
- Top-2 Accuracy: 95.41%
- Top-3 Accuracy: 98.47%

These results indicate that the model maintains strong predictive performance, even when considering top-k accuracy metrics. Furthermore, the results demonstrate that the proposed model achieves a good balance between accuracy and computational efficiency, making it suitable for real-time applications.

### 5.3.2 Accuracy and Loss Analysis

The training and validation accuracy curves provide insight into how effectively the model learns from the data. A consistent increase in accuracy indicates effective learning, while a small gap between training and validation accuracy suggests good generalization.

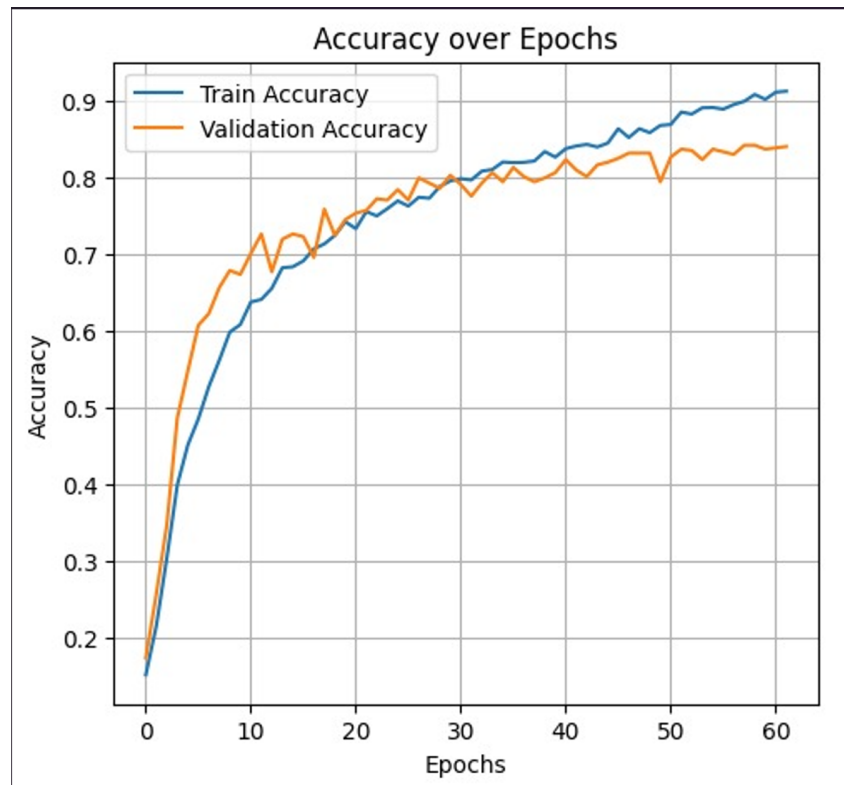


Figure 5.2: Training and Validation Accuracy Graph

The loss curves help in understanding how well the model minimizes prediction error. A decreasing trend in both training and validation loss indicates successful convergence.

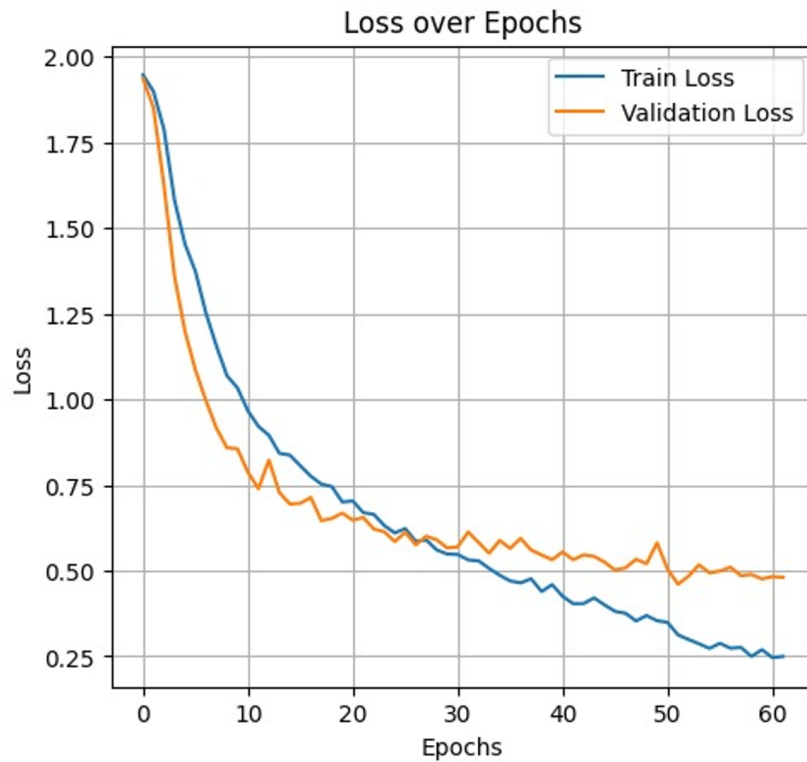


Figure 5.3: Training and Validation Loss Graph

### 5.3.3 Confusion Matrix Analysis

The confusion matrix provides a detailed visualization of classification performance across all emotion classes. It highlights correct predictions as well as misclassifications, helping identify challenging emotion categories.

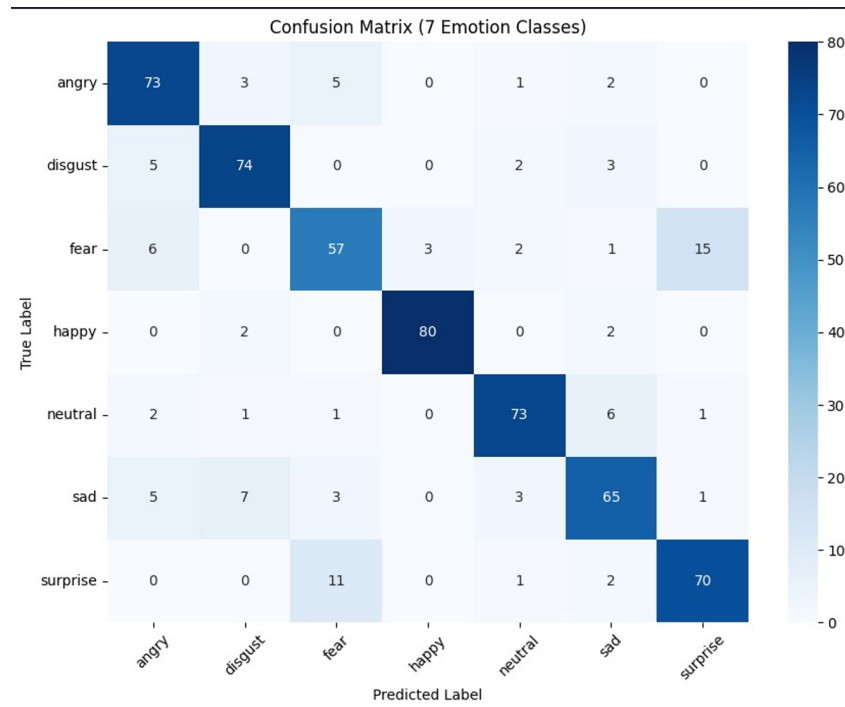


Figure 5.4: Confusion Matrix of Emotion Classification

Screenshots of the system during real-time emotion detection can also be included to demonstrate the practical implementation of the model.

### 5.3.4 Comparison with Existing State-of-the-Art Methods

The proposed system demonstrates competitive performance compared to traditional machine learning approaches due to the use of a custom CNN architecture and optimized preprocessing techniques. The model achieves a good balance between accuracy and computational efficiency, making it suitable for real-time applications.

### 5.3.5 Computational Complexity Analysis

The proposed model is computationally efficient due to the use of a lightweight custom CNN architecture. The model requires moderate computational resources for training and can perform real-time inference with minimal latency, making it suitable for deployment in practical environments.

### 5.3.6 Novelty of the Proposed Methodology

The novelty of the proposed system lies in its ability to perform real-time emotion recognition using a deep learning-based approach. The integration of face detection, image preprocessing, a custom Convolutional Neural Network (CNN) model, and real-time webcam implementation makes the system practical and efficient for real-world applications.

# Chapter 6

## Conclusion and Future Work

### 6.1 Summary of Work

The project titled “Real-Time Facial Emotion Recognition System using Machine Learning” focuses on developing an automated system to analyze and classify human emotions using facial expressions in real time. The system utilizes deep learning techniques along with computer vision methods to detect facial expressions and categorize them into seven emotion classes: angry, disgust, fear, happy, neutral, sad, and surprise.

The implementation involves preprocessing of image data using face detection and cropping techniques with OpenCV, followed by resizing and normalization. A custom Convolutional Neural Network (CNN) model is designed and trained to extract features and perform emotion classification. The model is trained using optimization techniques such as data augmentation, early stopping, and learning rate reduction to improve performance and prevent overfitting.

The system is further extended to support real-time emotion detection using webcam input, demonstrating its practical applicability in real-world scenarios.

### 6.2 Key Contributions

The key contributions of this project are as follows:

- Development of a deep learning-based emotion recognition system capable of classifying facial expressions into seven distinct emotion categories.
- Implementation of an efficient preprocessing pipeline including face detection, cropping, resizing, normalization, and data augmentation.
- Design and optimization of a lightweight custom CNN architecture for effective emotion classification.
- Integration of real-time emotion detection using OpenCV and webcam input, enabling practical deployment.
- Evaluation of model performance using accuracy metrics, top-k accuracy, and confusion matrix analysis.



## 6.3 Limitations of the Project

Despite achieving satisfactory performance, the proposed system has certain limitations:

- The model performance may be affected by variations in lighting conditions, facial orientation, and occlusions.
- The dataset used may not fully represent diverse real-world conditions, which can impact generalization.
- The system currently focuses on single-face detection and may not perform efficiently in multi-person scenarios.
- Real-time performance may vary depending on hardware capabilities.
- Similar emotions such as fear and surprise may sometimes be misclassified due to overlapping facial features.

## 6.4 Future Scope / Extensions

The project can be further improved and extended in the following ways:

- Enhancing real-time performance through model optimization and hardware acceleration.
- Extending the system to support multi-face detection for analyzing multiple individuals simultaneously.
- Incorporating larger and more diverse datasets to improve model robustness and accuracy.
- Integrating the system with intelligent applications for automated feedback and analytics.
- Developing a user-friendly web or mobile application for easier accessibility and deployment.
- Implementing emotion trend analysis to track behavioral patterns over time.

# Bibliography

- [1] I. Goodfellow, D. Erhan, P. L. Carrier, A. Courville, M. Mirza, B. Hamner, W. Cukierski, and Y. Bengio, “Challenges in Representation Learning: A Report on Facial Expression Recognition,” *arXiv preprint arXiv:1307.0414*, 2013.
- [2] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, “Gradient-Based Learning Applied to Document Recognition,” *Proceedings of the IEEE*, vol. 86, no. 11, pp. 2278–2324, 1998.
- [3] K. Simonyan and A. Zisserman, “Very Deep Convolutional Networks for Large-Scale Image Recognition,” *arXiv preprint arXiv:1409.1556*, 2014.
- [4] K. He, X. Zhang, S. Ren, and J. Sun, “Deep Residual Learning for Image Recognition,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.
- [5] E. Barsoum, C. Zhang, C. C. Ferrer, and Z. Zhang, “Training Deep Networks for Facial Expression Recognition with Crowd-Sourced Label Distribution,” in *Proceedings of the ACM International Conference on Multimodal Interaction (ICMI)*, 2016.
- [6] J. Whitehill, Z. Serpell, Y. C. Lin, A. Foster, and J. R. Movellan, “Toward Automatic Prediction of Student Engagement from Facial Expressions,” *IEEE Transactions on Affective Computing*, vol. 5, no. 2, pp. 201–216, 2014.
- [7] N. Bosch, S. D’Mello, and C. Mills, “Automatic Detection of Learner Engagement Using Video-Based Facial Expression Recognition,” in *International Conference on Educational Data Mining*, 2015.
- [8] F. Chollet, *Deep Learning with Python*. Manning Publications, 2018.
- [9] C. Shorten and T. M. Khoshgoftaar, “A Survey on Image Data Augmentation for Deep Learning,” *Journal of Big Data*, vol. 6, no. 60, 2019.